Smart Learning Environments

REVIEW

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Technologies and strategies in collaborative online international learning: systematic literature mapping

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Abstract

Collaborative Online International Learning (COIL) appears to be one of the potential responses to the world's educational needs. It emerged as an initiative to promote digital, collaborative internationalization, and its implementation and interest among teacher communities have steadily increased. This research analyzed COIL experiences, focusing on the technologies and strategies employed to contribute to educational innovation. A systematic literature mapping method was employed using the PRISMA protocol, identifying 66 articles from SCOPUS and WoS indexing databases. Analysis was conducted through categorization and classification guided by formulated secondary questions to consolidate the proposed mapping. Findings reveal that (a) COIL encompasses various design technologies and strategies, (b) no clear relationship is evident between the proposed strategy and technological mediation, (c) challenges include issues of connectivity, interaction, and digital literacy, (d) there is low integration of strategies proposing projects solving global challenges, and (e) potential exists for building virtual communities that address complex problems. This research adds value by highlighting the diverse approaches teachers and institutions have been undertaking and how technology might enhance and mediate the strategies utilized in these experiences.

Keywords: Collaborative online international learning, Higher education, Mapping studies, Educational innovation

Introduction

The need to address global issues indicated in the Sustainable Development Goals (SDGs) spotlights situations affecting the world, presenting opportunities educators have identified when proposing Collaborative Online International Learning (COIL) in their institutions. The State University of New York (SUNY) provides training to teachers, emphasizing the potential for COIL to strengthen essential 21st-century skills the workforce needs to address current challenges (SUNY Coil, 2023; SUNY Coil CENTER, 2013). Durand and Balhasan (2023) proposed a collaborative, multidisciplinary, multinational real-world project where students confront a problematic situation that they must resolve through collaborative and efficient communication. In this sense, COIL does not just facilitate a virtual exchange but emerges as a growing educational pedagogy thanks to a substantial body of research advocating for genuine integration of two or more courses from institutions that, geographically and culturally, may differ but join to develop projects in a shared area of interest (Rubin & Guth, 2022). Collaborative Online International Learning experiences offer many more benefits than merely strengthening or developing intercultural competencies, as they might initially be perceived.

These multilateral efforts among participating institutions must align synergistically among multidisciplinary teachers who motivate concerted efforts of students and gather the necessary elements to facilitate the collaboration and integration proposed in COIL. This becomes particularly relevant because research has identified several elements that must be considered, such as participant roles, the level of prior knowledge, the competency development required, collaboration platforms, learning preferences, proximity to the subject matter, the influential role of the teacher, and student-centered experiences (Kitjaroonchai & Loo, 2023; Naicker et al., 2021). This collaboration among educational institutions must address real-world challenges proposed by a company, as was done by Ballesteros-Sola and Magomedova (2023). The findings assert that implementing this multicultural pedagogy increased and positively impacted personal, social, and cognitive skills. These points show that many possible online collaboration experiences can promote learning that aligns with global realities.

Undoubtedly, technologies have benefited and enhanced these experiences, making it possible to conceive the classroom beyond its physical walls. The previous points are not new; the multitude of possible experiences using technology is unimaginable, as Owston (1997) expressed when referring to online educational resources, which are now utilized in various strategies, moments, and ways, bringing about changes in educational environments and experiences and mentioning that the Web could provide freedom to teaching and learning beyond physical and time barriers. Digital educational resources have supported various online teaching strategies, including COIL, dating back to the early 2000s (Rubin & Guth, 2022), but not conceived as they were viewed during the pandemic. Those strategies responded to all the COVID-19 restrictions and structured the new universal digital classroom, supported by video conferencing platforms still used to convene synchronous classrooms (Ota et al., 2023). Digital resources and strategies are now possible and valid but have different purposes than before. It can be asserted that digital resources are suitable for short educational interventions, content exposure, and limited experiences, whereas COIL is oriented toward generating deep and meaningful learning processes (MacNeill et al., 2014). In COIL, technology enables pedagogical learning experiences to meet learning expectations.

Some research on COIL have explored various elements related to student performance in these experiences, identifying necessary resources for implementation, while others focus on its innovative pedagogical character. Appiah-Kubi and Annan (2020) did a comparative study of a group of students participating in COIL who performed significantly better in project work due to intercultural collaboration and cooperation than a group that did not have the experience. Another study suggested that COIL should be supported by a methodology such as project-based learning to structure all phases, activities, and rubrics that apply to partner institutions and, thus, all students. It also recommended curating the ICT tools that must be used and forming project groups involving students from both institutions (Vasquez & Ramos, 2022). On the other hand, Balula (2022) suggested in the conclusions of the bibliometric review that four elements should be considered for further COIL studies: the commitment of the student, linguistic competency in the project development, the contents delivered for the development of activities, and the development or promotion of collaborative learning in these types of environments. These studies indicate the necessity to explore through systematic literature mapping the strategies, technologies, educational levels, and disciplinary areas in these proposed experiences to continue identifying constructs empirically that consolidate COIL as a new learning experience responding to the expectations of current education.

Previous literature review

Although there is little existing literature on COIL experiences, there has been a noticeable increase in publications over the last five years. This indicates an interest in the subject, which emerged during the Covid-19 pandemic but has continued, given that the year with the most publications to date is 2024. In this regard, the review by Fukkink et al. (2024) examined the pedagogical barriers and facilitators of intercultural learning in Collaborative Online International Learning (COIL) projects. To do so, it used the TPACK model as an analytical framework. This study concludes that, from a pedagogical perspective, it is essential to ensure collaborative work and encourage interactions between students, which enriches the process and constitutes a strategy for support and guidance. It also highlights the need for a vision of technology that goes beyond an instrumental or purely technical perspective. It should be noted that students propose new digital experiences to participate in these online platforms according to their preferences and knowledge, which contributes to these experiences. Finally, they recommend that future studies with more comprehensive data collection explore COIL more deeply, in order to identify which innovative methods best contribute to forming global citizens through this type of experience.

For their part, Sharma and Panackal (2025) acknowledge that research on COIL is still in its early stages and present a bibliometric study covering the years 2007–2024. The purpose of the study was to explore research gaps and identify emerging areas of inquiry within COIL. The study shows that the areas and disciplines that have implemented COIL are diverse, but it also highlights a trend in the healthcare sector and clarifies that it is still too early to see a significant impact on solving global health problems. However, it notes the potential of COILs, in which developed countries can work on global studies with developing countries according to their economy. This is where technologies, as mediators, would play an important role, as they should not be a barrier, but rather facilitate training through the use of appropriate online platforms. Finally, some challenges stand out, such as appropriate schedules, institutional support, collaboration, and teacher competence in this type of experience.

In line with the above, Shaw et al. (2025) focus on promoting intercultural competence among healthcare professionals by proposing a review that explores the implementation, effectiveness, and challenges of COILs in their training. According to their findings, COILs improve cultural competence, communication skills, and awareness of global health. They also highlight the challenges posed by technological barriers, schedule coordination, and curriculum alignment. In addition, they suggest using standardized assessment methods and long-term evaluations of the impact of COILs on education.

Research gap and study objectives

As can be seen, there is no literature review that has investigated the technologies used to mediate COILs and the pedagogical or didactic strategies on which they are based. It is clear that technologies are seen more as a challenge that requires digital skills and competencies in students and teachers to prevent them from becoming a barrier in these courses. However, it is clear that technologies have the potential to enrich these types of experiences, as they allow the creation of simulations or virtual environments that offer students immersive multicultural experiences that facilitate training, communication, collaboration, and online work to solve global problems.

Likewise, it is clear that collaboration is a cornerstone of COILs, and there is consensus on certain structural elements. These include the duration of the experience (ranging from 5 to 16 weeks), the integration of two groups from institutions in different countries, and prior preparation by teachers to ensure both curricular and technical coordination. However, it is not clear, and in several studies the pedagogical or didactic strategy that underpins this collaboration is not observed, so it is important to investigate these strategies that favor and enrich this type of experience, adapting them to the needs and structure of COILs.

For this reason, the objective of this mapping was to identify the technologies and strategies used in COIL experiences. To this end, seven research questions were posed, as indicated in Table 1. The first three questions aimed to provide a characterization and context. Questions four and seven focused on the type of study, a systematic mapping that explored the topics, findings, and skills reported as being fostered in this type of experience. Finally, questions five and six focused on the strategies and technologies.

Table 1 Research question and response categories

Research question (RQ)	Response categories
RQ1 What keywords do the author(s) identify in their studies, and how are they related?	Keywords, authors
RQ2 In which database is the published journal article indexed, in which year, and in which quartile?	WoS and Scopus, year of article publication, and quartile
RQ3 At what educational level was the experience developed, and which countries participated?	Primary, Secondary, Higher Education, and Continuing Education. Locations of the participants in the experience
RQ4 What were the themes and skills of the COIL experiences?	Keywords
RQ5 What technologies were used in the experience, and for what purpose?	Technology-based & Tools Platforms (Miranda et al., 2021)
RQ6 What were the pedagogical strategies used in the study?	Analysis strategies, reflection strategies, construction strategies, application strategies, collaboration strategies, and digital strategies (Ramírez-Montoya, 2022)
RQ7 What are the main findings (results) of the empirical studies?	Intercultural competencies, internationalization, COVID-19, interdisciplinarity, experiential learning, project-based learning, digital literacy, digital challenges, interculturality, global network learning, virtual communities, challenge-based learning, simulation of real company environments and contextual problems, and 21st-century competencies

Method

A Systematic Mapping Study (SMS) allows for the holistic identification of published evidence on a particular topic, providing an overview and detailed insight into what has been found, thereby discovering trends and gaps identified in the literature (Kitchenham & Charters, 2007). It is essential to clarify that a Systematic Literature Review (SLR) shares common elements with an SMS, such as the information search process and the selection or extraction of articles. However, they differ in their objectives. The SLR aims to synthesize and group evidence, while the SMS purports to identify areas and research trends (Petersen et al., 2015).

The mapping process involved two main phases: the first related to planning (Mapping planning) and the second (Mapping) to executing the planned activities. Each phase encompassed a series of activities in a systematic process in the proposed mapping (Kitchenham et al., 2010), as depicted in Fig. 1.

Research questions

Given the type of study, it was necessary to establish a broad objective and scope to identify and classify research trends in a specific area. Therefore, the objective was to uncover COIL experiences, emphasizing the technologies and strategies used, through secondary research questions that aimed to map those constructs and categories, enabling further investigations and reviews focused on specific elements of COIL environments (Petersen et al., 2008). Table 1 presents the formulated questions for the study.

Review protocol

The scope of the mapping led to the decision to use the PRISMA diagram (Moher et al., 2010), which allows for developing each activity of the mapping phase with the systematization and rigor required by this type of study, as evidenced in Fig. 2.

Search process

The search process was conducted similarly in both databases (Scopus and Web of Science) with similar search equations to retrieve a comprehensive construct on the

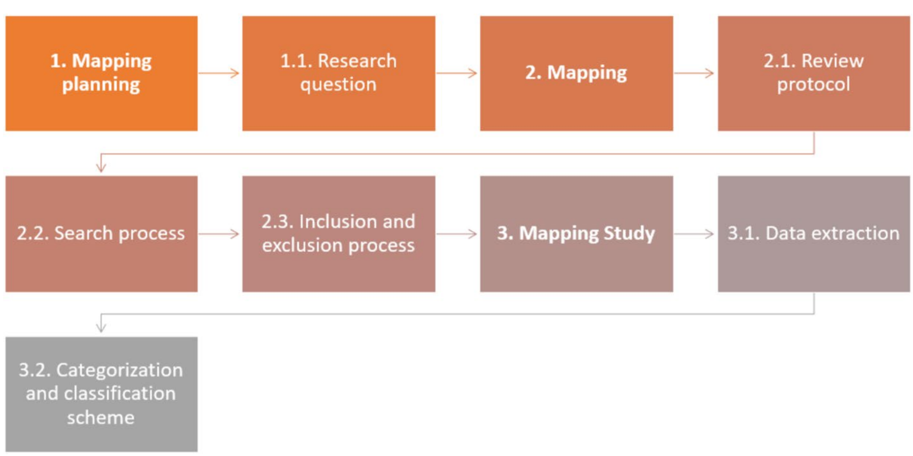

Fig. 1 Defined process to guide the methodological mapping (author's creation)

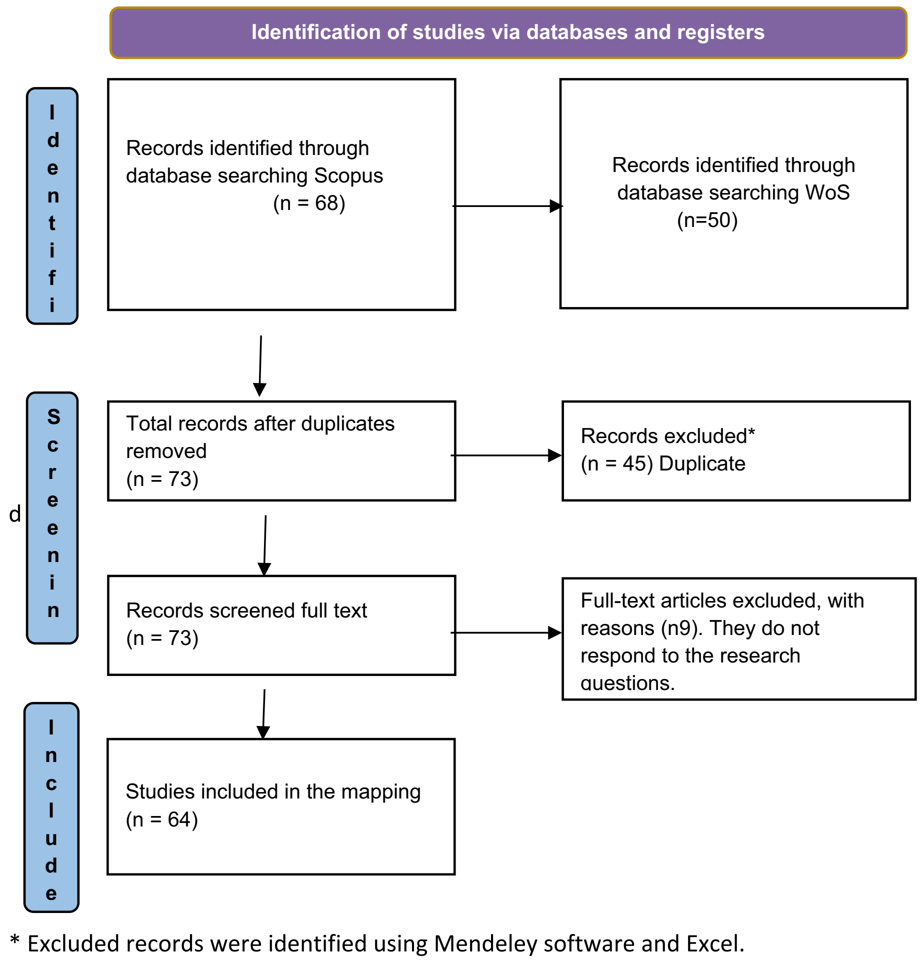

Fig. 2 Review protocol based on PRISMA

Table 2 Search equation

Search equation Scopus	Search equation WOS
TITLE-ABS-KEY ("collaborative online international learning") AND (LIMIT-TO (DOCTYPE,"ar") OR LIMIT-TO (DOCTYPE,"re"))	"collaborative online international learning"(All Fields). Refine: Articles and review

topic of interest. In the identification phase, exercises were conducted in both databases to observe the search results and their relevance. A complex equation was not necessary; it sufficed to search for the term within quotation marks to ensure that the concept was not divided into each of the words. The search was restricted to documents classified as articles or reviews, excluding all other document types. In the case of Scopus, the search was limited to the terms being present in the title, abstract, and keywords. The same equation was applied for Web of Science (WOS), but across all fields. Table 2 shows both search strings.

Inclusion and exclusion criteria

For mapping studies, it is important to clearly manifest the inclusion and exclusion criteria considered in the information search process (Kitchenham, 2004). The criteria considered for this study can be seen in Table 3, which includes the focus, document type, language, and publication date.

Data extraction process

The data extraction process was rigorous and followed what was outlined in the review protocol according to the PRISMA procedure, as demanded by these systematic mapping studies (Petersen et al., 2015). The key was to have a broad and general search equation that would include studies published on the topic but also sufficiently focused by including quotation marks to ensure they were studies referring to COIL experiences and limiting the search to articles and reviews. In the case of both databases, all results were included, which in this case were 68 from Scopus and 50 from WOS, for a total of 118. Then, 45 duplicate articles were removed in the extraction process, leaving 73 articles included.

Continuing with the selection process and ensuring that the articles were related to the research objective, it became necessary to read the entire study in some cases, following the recommendations of Kroll et al. (2018). It was found that nine articles were unrelated to COIL and excluded, leaving 64 articles for in-depth analysis. That validation was performed with the support of another researcher, following the suggestion of Brereton et al. (2007), who recommended that in systematic mappings, a researcher can be asked to verify the results and conduct an independent extraction, which can then be compared with the extraction already performed.

Finally, all the metadata was downloaded to an online Excel file to organize the included studies. Additionally, this allowed for condensing the information from the analysis of each quality question guiding the research. Database file https://goo.su/jeuC

Classification and categorization scheme

Seven research questions were proposed that required a classification as suggested by Petersen et al. (2008): four were closed-ended, with their answers primarily found in the title, abstract, or keywords of the articles or in their metadata; therefore, their encoding was structured and straightforward. Subsequently, two open-ended questions were found in which categories from other studies were established to synthesize and compile results to show trends. In the case of the technologies used, the classification proposed by Miranda et al. (2021) was utilized, as depicted in Table 4. The pedagogical strategies were based on Author (2022), as shown in Table 5. Question 7 was open-ended, lacking

Table 3 Inclusion and exclusion criteria

Inclusion criteria	Exclusion criteria
Studies on COIL experiences Collaborative online international learning Articles and reviews published in scientific Journals Articles and reviews in all languages Articles and reviews published at any date	Studies that its focus was not related to COIL Conference papers, books, book chapters, reports, working papers, or articles not published or indexed in Scopus or WoS

Table 4 Categories identified for technologies used

Technology used	What does it include?	Theoretical basis
Technology-based	IoT, AI, ML, CC, Cyber-physical systems, data science & Data Analytics, and Mixed reality	Current and emerging technologies that provide previously inconceivable capabilities based on techniques such as IoT, artificial intelligence, and learning machines to enrich platforms and tools in the education sector (Miranda et al., 2021)
Tools Platforms	Web conference platforms, Learning management systems, Collaborative virtual platforms, MOOCs, Remote and Cyberphysical labs, Robot teaching assistants, and hologram teachers	Tools and platforms used to enrich the teaching and learning processes according to the intention, allowing for the proposal of collaborative strategies, synchronous and asynchronous processes, complementary work, and support to different modalities (Miranda et al., 2021)

Table 5 Categories identified for pedagogical strategies (Classification based on Author, 2022)

Pedagogical strategy	What does it include?
Analysis strategies	Problems, debates, argumentation
Reflection strategies	Cases, metacognition, electronic portfolios
Construction strategies	Experience, games, projects
Application strategies	Challenges, evidence, research
Collaborative strategies	Open educational resources, massive online open courses, open innovation labs
Digital strategies	Robotics, augmented reality, virtual and remote laboratories

any predefined categorization or classification. It was identified after analyzing the reading, highlighting any findings or emphases that could be identified.

Results

To this point, the document analysis was managed through an Excel matrix that allowed for grouping all the information and conducting the process of encoding and categorization. The following link shares the data: https://doi.org/10.6084/m9.figshare.26308039.Below are the findings for each question.

RQ1. What keywords do the author(s) identify in their studies, and how are they related?

Of the 64 articles, two did not report keywords (Cotoman et al., 2021; Minei et al., 2021). The total number of keywords provided by the authors was 336. As Fig. 3 shows, eight main categories were identified that group and relate the article keywords, aiming to display the investigated trends. In this work, the highest-frequency category was"skills worked on or favored in these types of experiences,"such as collaboration, communication, critical thinking, and interculturality, among others, according to some of the recognized articles (Duffy et al., 2020; House et al., 2022; Simões et al., 2023; Suarez et al., 2020). In the"Focus"category, the topics or disciplines in which COIL was framed show a distribution across various areas of knowledge with a predominance in internationalization. Some examples include the articles identified with numbers (Garcia et al., 2023; Hackett et al., 2023; Marcillo-Gómez et al., 2016; Wood et al., 2022a, 2022b).

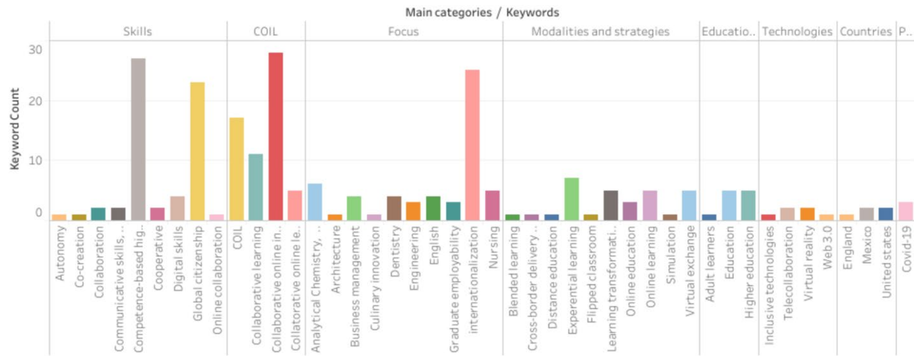

Fig. 3 Categories according to author keywords

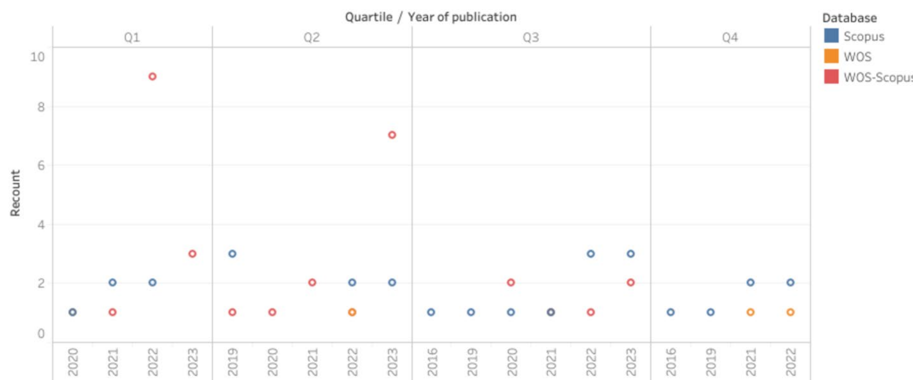

Fig. 4 Databases, quartiles, and publications years

The above data significantly contributed to the proposed mapping, highlighting several elements. The first was that this type of experience favors skills over specific topics or areas. Furthermore, it confirms that COIL is implemented in higher education, and although it was widely used during the pandemic, it was not merely an emergency pedagogy used at a specific moment; the modalities and implementation strategies varied, although there was a strong correlation with experiential learning.

RQ2. In which database is the published journal article indexed, in which year, and in which quartile?

Figure 4 shows that the first publication occurred in 2016 (Marcillo-Gómez et al., 2016), which is relatively recent, especially considering that no exclusion or filtering by year was applied within the search equation, addressing the question that guided the mapping.

The production was distributed across all quartiles, but with predominance in quartiles 1 and 2, which contained approximately 60% of the collected production, including some of the following articles (Banerjee et al., 2023; Ingram et al., 2021; Mestre-Segarra & Ruiz-Garrido, 2022; Quintana-Ordorika et al., 2023a, 2023b). Notably, in the last three years, 2021, 2022, and 2023, 40 articles were published out of 64 documents, equivalent to 62.5%.

With the onset of the COVID-19 pandemic, it is evident that COIL became an option to continue the internationalization of universities. It could be thought that COIL was positioned as a preventive and temporary measure. However, the COIL experiences were still ongoing and systematized in the last year, as evidenced by Hackett et al., (2023), Saftner & Ayebare, (2023); Naicker, 2023)]. In many cases, there was discussion of pedagogies that respond to the needs of contemporary society, considering the skills needed for a global understanding of problems, as seen in the article"Stepping up the game – meeting the needs of global business through virtual team projects" (Swartz & Shrivastava 2021), and others (identified as Duffy et al., 2020; House et al., 2022; Vicente et al. 2021). The journal with the most publications was Frontiers in Public Health (Collins et al., 2022; Wood et al., 2022a, 2022b; Case, et al., 2022), followed by journals related to education, business teaching, and engineering pedagogy (Hackett et al., 2023; Salmon et al., 2022; Membrillo-Hernández, et al., 2023).

RQ3. At what educational level was the experience developed, and which countries participated?

Forty-four countries had the experiences identified in the studies, demonstrating broad diversity and interest from various geographical areas. As Fig. 5 shows, the USA had a high percentage of participation, followed by Mexico and other countries in South America such as Brazil, Peru, and Colombia, and several European countries (Spain, Germany, and the United Kingdom, among others), and emerging participation from Asia and Africa. The total number of systematized experiences was 149, of which 98% were in higher education, one experience in a social organization (USA and Kenya) (Salmon et al., 2022), and one in secondary education (Australia and Japan) (Oakley, et al. 2023).

Although COIL projects were prominently developed in higher education to cultivate various skills related to interculturality and internationalization (Gray et al., 2021; Hildeblando et al., 2022; Marcillo-Gómez et al., 2016; Bragadóttir & Potter, 2019), there was an opportunity to cultivate other skills such as creativity, critical thinking, global citizenship, and digital competencies, which are not exclusive to higher education. Therefore, these practices can likely be migrated to secondary education. On the other hand, the identified social organization is remarkable, as it represents an ideal triangulation benefitting the dynamics of projects or challenges

Fig. 5 Frequency by country

addressed in these types of experiences. It correlates with companies and organizations that provide real-world problems to be solved by academia.

RQ4. What were the themes and skills of the COIL experiences?

Although COIL courses are typically articulated around a theme or content that facilitates alignment of professors, in many of these experiences, beyond content, the objective was to develop or work on a skill. In some cases, while studies focused primarily on disciplinary themes, they also addressed the skills that were enhanced (Inada, 2023; Romero-Rodríguez et al., 2022; Woodley et al., 2023; Watla-iad & Kradtap, 2022). In other instances, they only reported the impact on learning the subject (Gutiérrez-González et al., 2023; Kiegaldie et al., 2022a, 2022b; Fulco & Goldsmith, 2023) or highlighted the strengthening of a skill (Duffy et al., 2020; Hackett et al., 2023; Hildeblando et al., 2022; Naicker, 2023).

In the 64 studies available, 87 themes and skills could be identified. The most relevant themes were nursing, health, and business. The most predominant skills were in languages and interculturality, with communication and digital-related skills being less relevant. Figure 6 presents themes with the letter “T” and skills with the letter “S.” There were 55 studies referencing themes (63%) and 32 studies referencing skills (37%).

RQ5. What technology was used in the experience, and for what purpose?

RQ6. What were the pedagogical strategies used in the study?

Sixty-two strategies were identified in the 64 studies included in this mapping. In two articles, the strategy was unclear; although they referred to COIL, the experience was not described (De Klerk & Palmer 2022; Nethsinghe, et al., 2022). The majority of experiences implemented analysis strategies, in total 33, equivalent to 51%. These strategies involved debates that exposed issues, demonstrating the participants’argumentation. This highlights that COIL prioritizes discussions on topics covered during synchronous meetings, as evidenced by the following studies (Minei et al., 2021; Pouromid 2019; Anderson & Or, 2023; Bragadóttir, et al., 2019). The information presented so far is corroborated by the fact that the most commonly used technologies were video

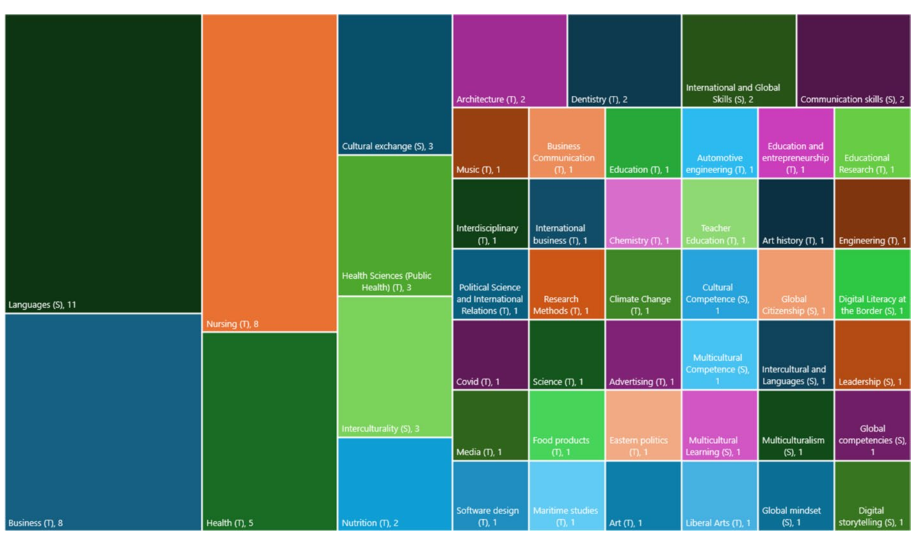

Fig. 6 COIL topics and skills

conferencing platforms and chat tools, without needing Learning Management Systems (LMS), collaboration platforms, or other tools primarily focusing on synchronous interactions according to Fig. 7. Similarly, the most commonly used technologies, regardless of the strategy, were video conferencing platforms and chat, according to Feng et al., (2021), Pouromid (2019), Saftner & Ayebare, (2023), and Swartz & Shrivastava, (2021).

On the other hand, fewer strategies were identified to address and propose solutions to general issues affecting a large part of the world by implementing inter-institutional projects and collaborations among different countries with diverse realities. However, some shared a comprehensive understanding of the problems (Mundel, 2020; Garcia et al., 2023; House et al., 2022). This can be seen in 8 experiences that logically used a combination of strategies in the proposed activities (Salmon et al., 2022; Mundel, 2020; Mackey & Aird, 2021; Suarez et al., 2020), such as construction strategies with reflection strategies, where project-based learning was combined with portfolio exercises to demonstrate progress and propose metacognitive exercises that help generate learning in students.

Significantly, there was no direct correlation between the strategy and technology, creating an unclear landscape. In construction and application strategies, in some cases, virtual collaborative platforms were used with video conferencing platforms, which would indicate a direct correlation with mediation technology. However, in other cases, only video conferencing and chat tools were used. In the case of digital strategies (Anderson & Or, 2023), a direct relationship with technology can indeed be observed, as simulators, virtual reality, and mixed reality were mentioned to carry out the proposed COIL.

RQ7. What are the main findings (results) of the empirical studies?

The purpose of this question was to identify in the literature the data that emerge in each study's results and conclusions, allowing for a broad understanding of COIL, its configurations over time, and possible recommendations for future experiences. In this regard, six categories and three levels of importance were determined based on their perspective and frequency, as shown in Fig. 8. In the first level of elements, topics related to the duration of experiences and COVID appeared. In the second level of categories was technology and its utility. Lastly, challenges and best practices were included in the order of degree of relevance.

The previous categorization indicated elements to think about when considering a COIL experience. In the categories of challenges and technology, very relevant findings

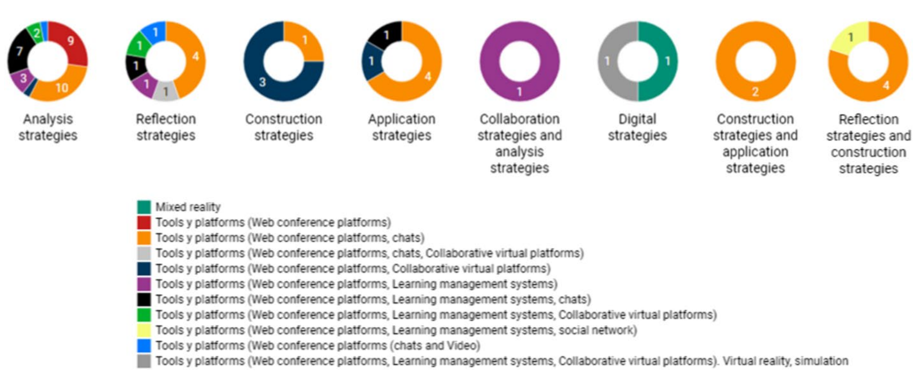

Fig. 7 Pedagogical strategies and technologies used

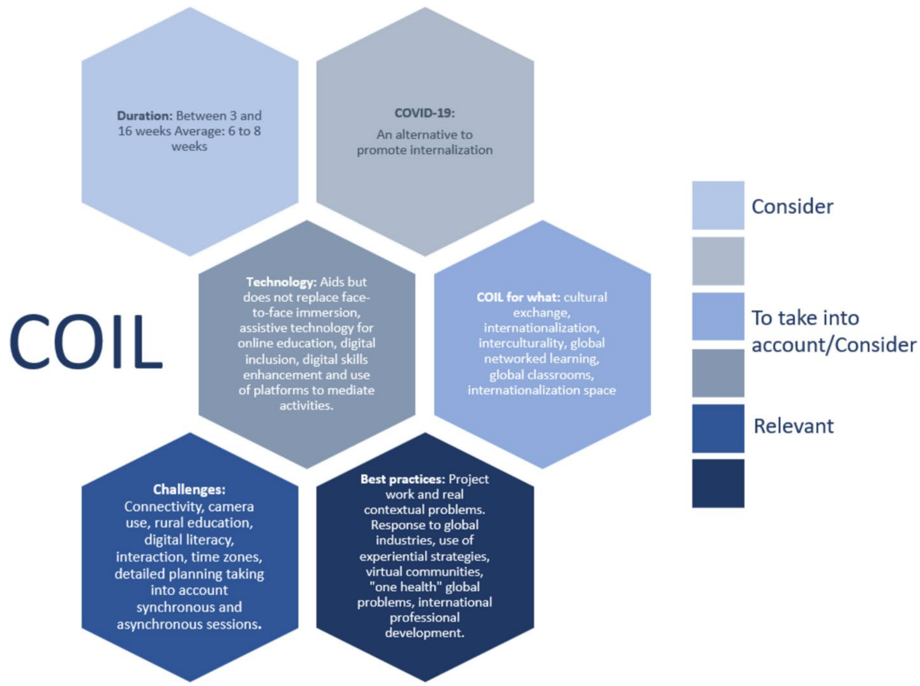

Fig. 8 Main findings and levels of significance

appeared related to connectivity issues, interaction among participants, aspects of digital literacy prerequisite to the experience to ensure the proper use of available technologies, and technological mediation for all proposed activities, both synchronous and asynchronous (Saftner & Ayebare, 2023; Wood et al., 2022a, 2022b; Naicker, 2023), and [20].

On the other hand, in the best practices category, it was confirmed that projects based on real-world issues are a compelling element for COIL; mentions included global classrooms or learning in a global network (Vicente et al., 2021; House et al., 2022). This opens up possibilities for interdisciplinary and inter-institutional work utilizing virtual communities to address global problems.

Discussion

COIL experiences are evolving spaces that, in recent years, have generated research opportunities. The COVID-19 pandemic accelerated their potential; the possibilities are currently broad and promising. Figure 3, which presents the keywords reported by the authors, evidences this. Most keywords were related to skills fostered by this type of experience and developed in these initiatives. Some authors, such as Woodley et al., (2023), de Castro et al. (2018), and Lohrová and Prošková (2021), affirmed that the experience was positive, considering the consolidation or contribution to the learning proposed, the strengthening of 21st-century skills for global education, training for work environments, employability, and the promotion of technology. Additionally, they highlighted the potential for addressing global challenges (Vicente et al., 2021). This is very important, as it reveals that COIL is not just an internationalization strategy; instead, it offers a much broader perspective as a strategy that responds to the current needs of education.

COIL experiences are still a field to be explored, studied, and systematized as they evolve and respond to some of the challenges of current education. It can be observed in high-impact publications that in 2016, only two articles were published in journals indexed in Scopus, in quartiles III and IV, respectively. Subsequently, there was a resurgence in publications in 2019 and 2021, showing momentum (see Fig. 4). Thus, COIL experiences have matured gradually, likely influenced by advances in connectivity, infrastructure, and technological elements, which continue to be a relevant factor today according to several studies (Anderson & Or, 2023; Quintana-Ordorika et al., 2023a, 2023b; Rubin & Guth, 2022; Vahed, 2021) (see Fig. 8). It would be relevant for COIL to retain the essence and purpose for which it was designed, not with a blurry and limited scope but with a view toward possible transformations, assuming its role as one of the most potent and relevant contextual pedagogical strategies.

The pandemic demonstrated that it is possible to create this type of online experience technologically. Of course, the issue is more complex than simply sharing a synchronous meeting between institutions; therein, the potential lies to be explored in the pedagogical innovations that can be designed and the mediating technological tools. Figure 7 shows that there is not always an alignment between the proposed pedagogical strategy and the technology used. Many studies revealed issues with access to tools proposed as mediation for these experiences (Saftner & Ayebare, 2023; Minnie et al., 2021). Students ended up using other tools, such as chat for interactions or other proposed activities, becoming a challenge to consider when proposing a COIL (Naicke, 2021; Niitsu, 2022). Some research showed evidence of alignment, such as digital strategies (sKiegaldie et al., 2022a, 2022b; Liu & Shirley, 2021), where virtual reality and mixed reality tools were used to respond to the proposed strategy. However, Kiegaldie et al. concluded that virtual reality immersion does not replace in-person experience (2022). The issue is not the replacement of in-person interactions; instead, it is crucial for educators to carefully select the technological tools they will use so that technology does not become an obstacle. Instead, it should facilitate the experience and provide possibilities previously unimaginable.

At this moment, COIL experiences are in configuration and development. It is necessary to conduct more iterations to unveil their myriad options, scopes, and limitations. Figure 8 suggests elements to consider, including recommendations regarding duration, challenges to overcome, and best practices. In this part, it is highlighted that in some COIL proposals, project-based work or experiential methodologies were employed, and real-world issues were addressed through partnerships with companies; this strategy can respond to education concerning trends such as global industries (House et al., 2022; Inada, 2023; Mundel, 2020; Salmon et al., 2022). The research studies can ensure positive results in student engagement, collaborative work, learning outcomes, and skills, such as communication, decision-making, and critical thinking, among others (Bauk & Fajardo-Flores, 2020; Cotoman et al., 2021; Swartz & Shrivastava, 2021; Wood et al., 2022a, 2022b). It is relevant to continue proposing COIL from both pedagogical and technological perspectives to enrich and repeat experiences that contribute to developing innovative educational processes and facilitate learning.

Conclusion

This research highlights the COIL strategies and technologies documented in the literature, as well as the various conceptions and transformations, but above all, it presents COIL as a field to explore and shape. There is no clear or single path for developing COIL experiences, although trends in their initial use and potential for the immediate future are evident. COIL has been and continues to be used to strengthen or fulfill internationalization in universities. Our mapping highlighted the experience of the COVID-19 pandemic. This situation undeniably forced universities to use COIL to continue internationalization processes, considering all the health restrictions imposed by governments that constrained mobility. It increased the interest in these experiences among teachers and institutions. However, in the current transition and evolution, COIL experiences are seen as necessary to address global issues, awaken student interest, strengthen soft skills needed for today's global job market, promote collaborative work, break down institutional and national barriers, and respond to educational needs of the twenty-first century.

On the other hand, there is no direct relationship between the proposed strategies and the technology used in these types of experiences. It could be assured that if the strategy includes the correct technology for mediation, it could have greater success, follow-up, and benefits for students and teachers. The above highlights that planning is one of the most relevant phases in the COIL process. It involves not only identifying an ally from another institution working on a similar topic and establishing contact but also creating a strategy using synchronous and asynchronous times, available technologies in both institutions, the participants' technological skills, curation of digital tools that aid and benefit the proposed work, timing, language, and schedules for the experience's development, allies, and companies. The contexts must realistically illustrate the problem or challenge being proposed and consider team formation, prior knowledge, skills to be developed recognitions, and assessment rubrics. All this goes beyond promoting internationalization; it involves working towards global networked learning and global classrooms that address worldwide issues.

When strategizing, project-based learning and experiential methodologies are compatible with COIL to offer more than just a remote connection with students from other universities or countries. The pandemic showed that COIL experiences are often possible from a technological standpoint. However, there is still much to explore that can be integrated into these experiences to enrich them. The value of this mapping's results is its revelations about what specific elements further studies should focus on for deeper exploration. Literature reviews can also continue documenting the progress of these experiences and their directions or trends.

One limitation is that the study focused on articles from scientific journals, excluding books, conference proceedings, and articles published outside Scopus and WoS. The excluded texts could provide a broader understanding of COIL in general. For future mapping, it may be meaningful to consider the other documents to achieve broader scope, coverage, and depth. Additionally, conducting systematic reviews focused on specific elements of COIL may be helpful.

Implications

In general terms, the study presented provides a better understanding of COILs. It systematizes the evolution of these learning experiences based on a compilation of research and broadens the perspective beyond conceiving COILs as an internationalization strategy to considering them a strategy in line with the needs of education in the digital age.

In practice, this raises the question of how to promote educational innovation by combining the pedagogical strategy that underpins the experience with the technology that enhances and enriches it. This is the most important aspect, since, although COIL courses have a clear structure, it is essential to define or investigate the strategies that are most closely linked to these experiences in order to continue developing experiences based on pedagogy and not only on internationalization as the ultimate goal. In addition to considering technology as a tool that enhances and enriches the experience, not simply as a means of mediation.

This is relevant for decision-making, as the study presents the current state of COILs, but above all highlights their potential to respond to global issues, in line with the Sustainable Development Goals. In this way, COIL experiences could become an innovative learning strategy in line with the educational challenges of the digital age.

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Author contributions
HR and MR planned the SLR following the PRISMA protocol. Together they developed the data analysis process working collaboratively on the matrix used for this purpose, which is shared in the same document. The writing of the article was carried out between the two of them, in some sections as main writer and in others as secondary author.

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Availability of data and materials
The datasets generated and/or analysed during the current study are available in https://goo.su/jeuC

Declarations

Competing interests
The authors declare that they have no competing interests.

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